

Solution Requirements

Phase 2: Requirement Analysis

shopEZ E-commerce Marketplace | MERN Stack | June 2026

1. Functional Requirements (FR)

The system must deliver the following capabilities, structured by user access level via Role-Based Access Control (RBAC).

FR-1: User and Authentication Services

ID	Requirement
FR-1.1	Standard users must be able to register an account with a unique email, name, and hashed password.
FR-1.2	Users must be able to log in securely to retrieve a signed JSON Web Token (JWT) session.
FR-1.3	The system must restrict administrative dashboard routes to users holding the admin or seller roles.

FR-1.4	Standard users must be able to manage an address book containing multiple shipping locations, selecting a default for checkouts.
--------	--

FR-2: Product Catalog and Discovery

ID	Requirement
FR-2.1	The database must list products displaying name, description, price, stock inventory, categories, and images.
FR-2.2	The frontend must provide full-text search with live autocomplete suggestions.
FR-2.3	The interface must support dynamic filters based on Category, Price Range, Minimum Ratings, and Stock Availability.
FR-2.4	The frontend must support sorting: Price (Low to High, High to Low), Ratings, and Release Date (Newest).

FR-3: Shopping Cart and Wishlist Lifecycle

ID	Requirement
FR-3.1	The system must offer a persistent shopping cart linked to the user context.
FR-3.2	Users must be able to modify item quantities in the cart, subject to real-time stock limits.
FR-3.3	The system must maintain a saved Wishlist, permitting users to move favorites to cart in a single click.

FR-4: Checkout, Coupons and Payments

ID	Requirement
FR-4.1	The system must guide the customer through a multi-step checkout wizard (Shipping Address, Shipping Speed, Coupon Apply, Mock Payment, Order Confirmation).
FR-4.2	The coupon engine must validate active coupon codes, applying percentage or fixed discounts and recalculating itemized taxes and shipping fees.
FR-4.3	The checkout must communicate with a mock payment intent endpoint to securely complete transactions.

FR-5: Administrative Operations

ID	Requirement
FR-5.1	Admins and sellers must hold complete CRUD privileges over products.
FR-5.2	Sellers and admins must be able to upload product images, stored locally via Multer.
FR-5.3	Admins must be able to modify order fulfillment milestones (Placed, Confirmed, Packed, Shipped, Out for Delivery, Delivered).
FR-5.4	Admins must be able to block/unblock customer accounts, instantly revoking active sessions.
FR-5.5	Admins must be able to manage coupon campaigns (create, toggle active state, delete).

2. Non-Functional Requirements (NFR)

NFR-1: Security and Protection

ID	Requirement
NFR-1.1	Passwords must be hashed using the Bcrypt algorithm (salt round factor 10) before database commits.
NFR-1.2	Session JSON Web Tokens must be transmitted via HTTP-Only, Secure cookies to prevent XSS-based token extraction.
NFR-1.3	Enforce strict API middleware validation to deny unauthenticated access to protected routes.

NFR-2: Performance and Optimization

ID	Requirement
NFR-2.1	The UI must maintain high responsiveness utilizing React's virtual DOM to update components without full-page reloads.
NFR-2.2	Product images must be served via Express static file serving from /backend/uploads for quick local access.
NFR-2.3	API responses for the catalog must complete within 200ms under standard loads.

NFR-3: Reliability and Integrity

ID	Requirement
NFR-3.1	Database commits during checkout must ensure atomic stock deductions to prevent oversells.

NFR-3.2	If the primary MongoDB cluster is unavailable, the backend seeder must automatically instantiate a local In-Memory MongoDB fallback.
---------	--

NFR-4: Usability and Accessibility

ID	Requirement
NFR-4.1	The user interface must utilize a fully responsive layout that fits mobile, tablet, and desktop viewports.
NFR-4.2	Design styling must utilize Outfit or Inter fonts with clear typography hierarchies.